

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU is text-only with explicit exclusion of reading-comprehension passages, image-based items, and >4-option items **[Q33]**, plus INDICLID + Unicode-based language filtering **[Q34]**, duplicate removal **[Q35]**, and multi-stage manual cleaning **[Q31, Q32, Q36]**. This matches the deployment’s text-only mobile interface and standard Devanagari script with no RTL/capitalization concerns. Observed Hindi items render in standard Devanagari with minimal Roman intrusion **[DATASET-D12, D21]**. One residual concern: dataset analysis observed at least two malformed items with truncated stems across language partitions **[DATASET-D34, D35]**, suggesting filtering imperfections.

ID	Evidence
Q31	“Despite our best efforts to maintain the quality of questions collected, some amount of noise or errors may still be present. To address potential noise in the questions, we employ multiple layers of manual and automated cleaning filters.” (p.4)
Q32	“Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise.” (p.4)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q34	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q35	“To further refine the dataset, we remove any duplicate questions to retain only the unique ones.” (p.4)
Q36	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D12	रेडक्लफि रेखा भारत की सीमाओं को नमिन्लखिति में से कसि देश के साथ वभिजति करती है? ... पाकस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D21	एक भौगोलिक क्षेत्र में नमिन्लखिति वशिष्ट वशिषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — “brought a” with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിൽപ്പം — Incomplete stem (“Directions: marble can be used”) — same truncated structure appears across languages

Strengths

- Text-only MCQ format directly matches deployment input modality.
- Devanagari script handling is well-supported **[Q34]**, no signal-distribution mismatch for Hindi.
- Multi-layer cleaning pipeline **[Q31, Q32, Q36]** reduces noise.
- Observed code-mixing in Hindi items remains below the ~10% ceiling **[DATASET-D12, D21]**.

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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution matches: text-only Devanagari [Q33, Q34] aligns with the deployment's mobile text interface. No image/audio modality mismatch.

IF-2: Regional infrastructure (Android-dominant smartphones, Devanagari Unicode standard) supports text-only MCQ ingestion. Bandwidth-sensitive but format-compatible [WEB-6].

IF-3: Domain-specific form: MCQ-with-4-options matches central-exam Prelims pattern (UPSC, SSC, banking are MCQ-based). Reading-comprehension exclusion [Q33] removes one common exam format but is acceptable for the deployment's objective-prep scope.

IF-4: Minor form mismatches: (a) some items have truncated stems suggesting reading-comprehension remnants [DATASET-D34, D35]; (b) reading-comprehension exclusion means RC-style UPSC CSAT questions are not represented.

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WEB-6	TRAJ data shows internet penetration patchy in Bihar — relevant to deployment connectivity assumptions (https://trai.gov.in/sites/default/files/2024-11/Cons_P_14092023.pdf)

Information Gaps

- Frequency of malformed/truncated items in the full Hindi test split is unknown.